

Ticket Master

functional requirements:

user should be able to book a ticket based on his preference,
user should be able to search for an event based on his preference,

Non-functional requirements:

Data-partitioning

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency.

high throughput ,

CAP theorem (Here we need to prioritise strong consistency for booking scenario in order to avoid Double booking or double payment & for search scenario we can prioritise 99.99% uptime i.e Availability>>Consistency)

Core Entities:

User,
booking,
history,
payment

API :

POST:/booking

```
{
  userID,
  bookingStatus
}
```

GET:/locations/{locationID}

POST:/payment

```
{
  paymentID,
  transactionStatus
}
```

HLD :

